WORKED SOLUTION

STEP III 1989, Question 16

Probability and Statistics • Elementary probability

STEP III 1989, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s3-q16

The key idea

Three yes/no characteristics split the population into eight cells, and the eight statements are just linear equations for the proportions in those cells. Fill in the green-eyed cells first (the conditions there are the most specific), then the hazel-eyed ones. The statistical parts then rest on one number, the probability that a Ruritanian is narrow-headed, and ask whether the observations are surprising under that value.

Solution

The printed paper has three slips of the pen. We read "arrow-headed" in (iv) as *narrow-headed*, "fair-headed" in (vi) as *fair-haired*, and "hazel-haired" in (viii) as *hazel-eyed*. These are the only sensible readings, since the three characteristics are hair (fair or dark), eyes (green or hazel) and head (narrow or broad).

The proportion of narrow-heads

Choose a Ruritanian at random. Write F, D for fair- and dark-haired, G, H for green- and hazel-eyed, and N, B for narrow- and broad-headed. We are told $\mathrm{P}(F) = \frac{1}{4}$ and $\mathrm{P}(G) = \frac{1}{5}$.

Green-eyed cells. By (v), $\mathrm{P}(N \mid G) = \mathrm{P}(B \mid G) = \frac{1}{2}$, so

$$\mathrm{P}(G \cap N) = \mathrm{P}(G \cap B) = \tfrac{1}{10}.$$

By (vi), the green-eyed broad-heads split equally between fair and dark hair:

$$\mathrm{P}(F \cap G \cap B) = \mathrm{P}(D \cap G \cap B) = \tfrac{1}{20}.$$

By (iv), $\mathrm{P}(F \cap G \cap N) = 0$, so $\mathrm{P}(D \cap G \cap N) = \frac{1}{10}$.

Hazel-eyed cells. Let

$$a = \mathrm{P}(F \cap H \cap B), \quad b = \mathrm{P}(F \cap H \cap N), \quad c = \mathrm{P}(D \cap H \cap B), \quad d = \mathrm{P}(D \cap H \cap N).$$

- Fair hair: the green-eyed fair-haired make up $0 + \frac{1}{20}$, so $a + b = \frac{1}{4} - \frac{1}{20} = \frac{1}{5}$.
- Hazel eyes: $a + b + c + d = \frac{4}{5}$, so $c + d = \frac{3}{5}$.
- Statement (vii): $\mathrm{P}(B \cap D) = \frac{1}{20} + c = \frac{1}{2}$, so $c = \frac{9}{20}$ and hence $d = \frac{3}{20}$.
- Statement (viii): $\mathrm{P}(F \cap B \mid H) = \mathrm{P}(D \cap N \mid H)$; multiplying by $\mathrm{P}(H)$ gives $a = d = \frac{3}{20}$, and so $b = \frac{1}{20}$.

All eight proportions, in twentieths, are:

	$G \cap N$	$G \cap B$	$H \cap N$	$H \cap B$
F	0	1	1	3
D	2	1	3	9

They are all non-negative and add to 20, so the model is consistent. Adding the narrow-headed columns,

$$\mathrm{P}(N) = \frac{0+2+1+3}{20} = \frac{3}{10}.$$

So the model says that 30% of Ruritanians are narrow-headed.

Six broad-headed acquaintances

If my six acquaintances were a random sample, independent of one another, the chance that all six are broad-headed would be

$$\left(\tfrac{7}{10}\right)^6 = 0.1176\ldots \approx 0.12.$$

Ask yourself Is a 12% event good evidence against the model?

No. Something with probability about 1 in 8 is not unusual; it would not be significant at any conventional level such as 5%. The data are also weak in a second way: one's acquaintances are not a random sample. They may share a family, region or social group, so they need not be independent, and they need not represent the whole population. The observation gives at most very slight grounds for doubt.

A random sample of 200

Let X be the number of narrow-heads in a random sample of 200. The population is large, so the draws are effectively independent and, under the model, $X \sim \mathrm{B}(200, \frac{3}{10})$, with

$$\mathrm{E}(X) = 60, \qquad \mathrm{Var}(X) = 200 \times 0.3 \times 0.7 = 42, \qquad \sqrt{42} \approx 6.481.$$

Since np and $n(1-p)$ are both large, we use the normal approximation $X \approx \mathrm{N}(60, 42)$, with a continuity correction:

$$\mathrm{P}(X \le 50) \approx \Phi\left(\frac{50.5 - 60}{\sqrt{42}}\right) = \Phi(-1.466) \approx 0.071.$$

So under the model, 50 or fewer narrow-heads would happen about 7% of the time.

Ask yourself Should we now reject the model?

Not at the 5% level: a one-tailed test of $p = 0.3$ against $p < 0.3$ gives a p-value of about 0.07. The result does cast some doubt on the model, though. The sample proportion $\frac{50}{200} = 0.25$ is about 1.5 standard deviations below the predicted value. It also points the same way as my acquaintances, who had fewer narrow-heads than expected. It would be sensible to collect more data before trusting the model's figure of 30%. (If the test were two-tailed, the p-value would be about 0.14, and the evidence weaker still.)

Checks

- Each of the eight conditions can be read straight off the table, for example (vii): $1 + 9 = 10$ twentieths $= \frac{1}{2}$.
- A computer check confirms that the cell proportions are the unique solution, that $\mathrm{P}(N) = \frac{3}{10}$ and that $(0.7)^6 = 0.117649$. The exact binomial probability $\mathrm{P}(X \leq 50) = 0.0695$ is close to the normal approximation 0.0713.

Common slips

- Reading (v) or (vi) as statements about the whole population rather than conditional ones: "as likely as" within the green-eyed group means equal halves of $\frac{1}{5}$.
- Forgetting the continuity correction. Using 50 instead of 50.5 gives $\Phi(-1.543) \approx 0.061$.
- Treating acquaintances as a random sample without comment.

Beyond STEP

The last part is a hypothesis test, and the two pieces of evidence raise a natural question: how should several weak pieces of evidence be combined? Undergraduate statistics answers this with likelihood (and, for the Bayesian, posterior probabilities), which measure how well each value of p explains *all* the data together. Grimmett and Welsh, *Probability: An Introduction*, gives the probability background, and Blitzstein and Hwang, *Introduction to Probability*, has an approachable account of the Bayesian view.

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